AZURE + FINOPS REFERENCE

Industry Best Practices: Storage, Managed Disks, and Data Services

Optimize data cost through lifecycle, tiering, capacity, retention, and access-pattern decisions. Treat data durability, recovery, legal hold, and compliance obligations as first-class constraints.

Applicable resource types

Microsoft.Compute/disks, Microsoft.Storage/storageAccounts, Microsoft.Sql/servers, and associated backup, snapshot, and data-retention resources.

Cost optimization practices

Match redundancy, performance tier, and access tier to documented recovery and latency requirements. Use lifecycle management for eligible data after classifying access patterns and retention obligations.

• Review disk size, performance tier, attachment state, IOPS/throughput demand, snapshots, and backup dependencies together.

• Use storage analytics to identify suitable candidates for tiering and lifecycle policy, scoped to approved containers or prefixes.

• For databases, use measured CPU/DTU/vCore, storage, IO, concurrency, and performance baselines before changing service tier.

• Prefer capacity and rate commitments only after workload usage is stable and waste has been addressed.

Operational safeguards

Unattached or cold data is not automatically disposable. Validate ownership, restore requirements, legal hold, backup chain, retention policy, and business continuity requirements.

• Apply lifecycle policies in a staged scope and verify restore behavior.

• Do not reduce redundancy, backup retention, database tier, or recovery objectives without owner and risk approval.

• Keep an auditable decision record for data-classification and retention exceptions.

Evidence and approval model

Decision area	Evidence to collect	Safe action gate
Storage lifecycle	Access frequency, prefix/container ownership, retention and legal requirements	Stage policy; validate restore and exception workflow
Managed disk tier	IOPS, throughput, capacity, attachment state, snapshots	Confirm workload and recovery dependency; plan rollback
Database tier	Peak CPU/DTU/vCore, IO latency, storage and workload SLO	Load test and approve maintenance/change window

Source material

This reference summarizes the following official guidance. Check the linked sources for current service limits, billing terms, and implementation details.

- **Azure Well-Architected data cost strategies**
https://learn.microsoft.com/en-us/azure/well-architected/cost-optimization/optimize-data-costs

- **Azure storage architecture guidance**
https://learn.microsoft.com/en-us/azure/architecture/storage/storage-get-started

- **Azure Cost Management recommendations**
https://learn.microsoft.com/en-us/azure/cost-management-billing/costs/tutorial-acm-opt-recommendations